ASR-Roundtrip Evaluation Can Mask Context- and Convention-Dependent Reading Errors in Chinese News TTS

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Abstract

Automatic speech recognition (ASR) roundtrip evaluation is widely used as a scalable proxy for text-to-speech (TTS) intelligibility. We show that it can miss reading errors perceived by listeners when the correct spoken form depends on discourse context or domain convention, as in sports scores, aircraft models, technical units, and membership names. TTS may produce a fluent but wrong reading while ASR returns the intended or surface-correct text. We formalize this false-negative event and build a frozen 200-case reading-risk benchmark from a source pool of 108,124 company-produced Chinese news scripts used in a production TTS workflow and a 5K synthetic hard-case pool. In a targeted audit of 110 high-risk MiMo TTS cases, 46 are confirmed masked, 9 contain errors exposed by ASR, and 55 contain no Raw TTS error. Context isolation re-exposes local wrong-reading evidence in 18/46 masked cases. A Raw-only CosyVoice audit on the same candidates confirms 51 masked cases. We then test the 97 files labeled confirmed masked across the two TTS audits with two open-source ASR systems: Qwen3-ASR-1.7B surface-recovers 40, whereas Paraformer recovers 2. Among Qwen's 19 full-context recoveries on MiMo audio, aligned-span transcription changes 12 to a wrong or noncanonical form. The results establish a reproducible but evaluator-dependent TTS-wrong/ASR-surface-correct failure mode and support using ASR roundtrip for screening rather than as standalone ground truth for Chinese news reading-risk evaluation.

Keywords: text-to-speech evaluation, ASR roundtrip, Chinese news TTS, context-dependent reading, text normalization, human listening audit

1 Introduction

ASR roundtrip evaluation offers a convenient way to evaluate TTS at scale: synthesize a reference text, transcribe the generated speech, and compare the transcript with the reference or an expected spoken form. This procedure is inexpensive, repeatable, and often informative about intelligibility. Its reliability, however, depends on the ASR system and evaluation protocol [1, 2]. An ASR transcript is an inference conditioned on acoustics, language priors, normalization behavior, and decoding conventions; it is not a transparent record of everything a listener hears.

Chinese news makes this distinction consequential. A score such as *13-11* should be read as 十三比十一 in a sports report, but TTS may produce the fluent range-like reading 十三至十一. An aircraft model such as 伊尔-76 should not be read as a negative number. Strings such as *640kW*, *350Wh/kg*, and *88VIP* require technical or product conventions. The resulting errors can remain acoustically fluent and locally plausible. If ASR then transcribes the wrong audio as *13-11*, 伊尔-76, or a normalized unit string, a transcript-based evaluator can score the case as correct even though a listener hears the error.

We call the central event a **masked false negative**: TTS produces a wrong target-span reading, but an ASR route returns the intended or surface-correct text. The failure is narrower than the familiar observation that ASR metrics do not perfectly match human judgments. It identifies a concrete direction of error in which a seemingly successful transcript hides a listener-perceived TTS defect. Figure 1 summarizes the chain.

We use *Context-Dependent Reading Decisions* (CDRD) for written spans whose correct reading depends on context beyond the local character sequence. We track CDRD-entity and CDRD-polyphone cases separately and include CDRD-adjacent convention-dependent risks, such as unit and membership strings, when they expose the same evaluator failure. This boundary is important: sports scores and model names mainly illustrate contextual recovery, while units and membership names can

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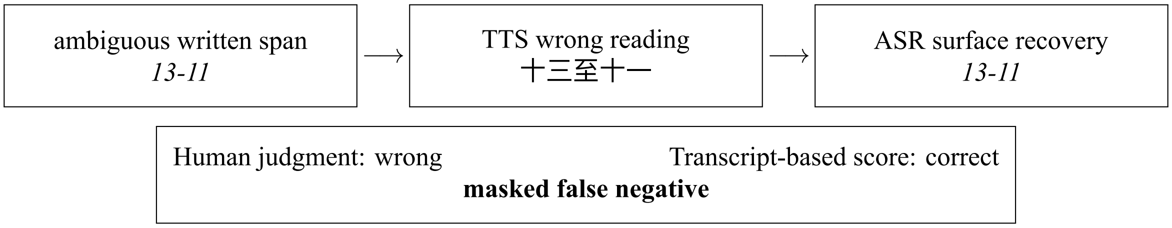

Figure 1: The evaluator-failure chain studied in this work. The transcript can look correct even when the target-span reading is wrong.

involve ASR canonicalization or text normalization. Both can create false negatives, but they need not arise from an identical decoder mechanism.

Our evidence combines a frozen reading-risk benchmark, human listening, a targeted audit with all outcomes reported, context isolation, and cross-system controls. The main contributions are:

- **A specific evaluator-failure formulation.** We formalize and audit TTS-wrong/ASR-surface-correct false negatives for Chinese news reading risks.
- **Human evidence with a complete outcome accounting.** A 110-case targeted MiMo audit yields 46 confirmed masked cases, 9 exposed errors, and 55 no-error cases.
- **Mechanism evidence.** MiMo span isolation re-exposes 18/46 cases, and a protocol-matched Qwen comparison changes 12/19 full-context surface recoveries to wrong or noncanonical forms on aligned spans.
- **Cross-system controls.** A CosyVoice audit reproduces the failure family, while Qwen3-ASR and Paraformer show sharply different surface-recovery behavior on the same human-confirmed wrong readings.
- **Human-anchored evaluation resources.** We document audio-first annotation, independent agreement checks, risk-span scoring, protocol sensitivity, and reproducibility materials.

The targeted samples establish the failure mode rather than its natural incidence. A rule-instantiated Structured condition is retained as an oracle-style diagnostic of explicit reading decisions, not as a deployable TTS frontend.

2 Related Work

2.1 Text normalization and Chinese reading decisions

Text normalization converts non-standard written forms such as numbers, dates, abbreviations, and units into speakable sequences. Classical systems use grammars and recoverable intermediate representations [3, 4]; later work uses minimally supervised and neural models [5, 6]. Chinese text normalization is especially exposed to mixed Chinese characters, Arabic digits, Latin letters, and symbols, motivating rule-guided neural models [7]. Recent LLM-based work also treats normalization as a context-sensitive generation task [8].

CDRD overlaps with text normalization but emphasizes a different boundary. The same hyphenated surface pattern may encode a score, a range, a model, or a subtraction-like expression; token shape alone is insufficient. Likewise, Chinese grapheme-to-phoneme systems use context for polyphone disambiguation [9, 10, 11, 12, 13, 14]. We treat classical polyphone ambiguity as one reading-risk category but separate it from mixed-script entity and convention decisions.

2.2 ASR-based and learned TTS evaluation

ASR-derived measures are widely used as scalable intelligibility proxies. Prior work has studied confidence intervals for ASR-based TTS evaluation and warned that ASR error rates do not necessarily predict human utility [1, 2]. Modern systems such as Whisper are robust transcribers trained with large-scale weak supervision [15], but robust transcription can also introduce strong linguistic and formatting priors.

Learned speech evaluators target quality, naturalness, intelligibility, or broad pronunciation. Representative systems include MOSNet [16], NISQA [17], and VoiceMOS [18]. SpeechLMScore [19] and TTS-PRISM [20] extend this line with speech-language-model signals. These tools are valuable, but a fluent domain-specific wrong reading may not strongly affect a generic quality score. Our contribution is not a new general metric. We document a false-negative mechanism and show why human-anchored span-level evaluation remains necessary for the targeted risk family.

3 Reading-Risk Formulation

3.1 CDRD and adjacent risks

Let x denote a written target span and c its surrounding context. Let $r^*(x, c)$ be the intended reading under that context. A span is a CDRD when $r^*(x, c)$ cannot be reliably determined from x alone, or when there exist contexts c_1 and c_2 for which the appropriate readings differ. Entity identity, discourse role, and domain convention can all contribute.

We use three case-level labels:

- **CDRD-entity**: scores, military or aircraft models, financial quarters, generation labels, and entity-dependent hyphenated forms;
- **CDRD-polyphone**: Chinese character pronunciation ambiguity, tracked separately because it is closer to conventional G2P;
- **CDRD-adjacent**: technical units, percentages, mixed-script strings, memberships, abbreviations, and foreign names that are convention-dependent or vulnerable to ASR canonicalization.

Figure 2 separates the strict CDRD family from adjacent risks. The adjacent set is included because it exhibits the same evaluator-level outcome, not because every unit or membership error satisfies the strict contextual definition.

Chinese news reading risks

Strict CDRD
context-dependent reading decisions

CDRD-entity
• sports scores and formations: *13-11*, *4-3-3*
• aircraft or military models: 伊尔-76, *F-35*
• generation labels and context-dependent hyphens

CDRD-polyphone
• context resolves character-level pronunciation

CDRD-adjacent
convention-dependent and ASR-maskable risks

• technical units and symbols: *N·m*, *Wh/kg*, *kW*, *V*
• Latin-letter strings and abbreviations: *AI*, *GDP*, *TOPS*
• membership and product forms: *88VIP*
• other conventional readings that need not satisfy the strict contextual definition

Included when the same TTS-wrong/ASR-surface-correct evaluator failure is observable.

Figure 2: Taxonomy and claim boundary for Chinese news reading risks. The two branches can share an evaluator-level failure without implying an identical decoder mechanism.

Each case is annotated at the risk-span level with the written form, risk type, expected reading, known negative readings when available, context evidence, and whether transcript-based scoring is permitted. Polyphone and foreign-name spans can be human-only because an ASR transcript may preserve the same characters despite a wrong pronunciation.

Accepted readings may form a set when a domain permits established variants. For the aircraft model *737-8*, public sources document both 七三七减八 and 七三七杠八, with 减 described as the industry-internal convention [21, 22]. We accept both variants; 七三七负八 and the range-like 七百三十七到八 remain errors. This post-freeze clarification leaves the 200-case human scores and the 110-case audit outcomes unchanged, but it changes one aligned-isolation label from exposed to still masked.

3.2 Masked false negatives

Let $y = T(t)$ be audio synthesized from text t. For a target span x, define $H(y, x) = 1$ when human listening judges its reading correct and $A(y, x) = 1$ when an ASR-backed scoring route returns the written target or an accepted expected form without exposing a known wrong form. A masked false negative occurs when

$$H(y,x) = 0 \quad \wedge \quad A(y,x) = 1. \tag{1}$$

Equation 1 deliberately separates human audio correctness from transcript recovery. $A(y,x) = 1$ does not imply that the acoustics contain the intended reading; it records the evaluator's surface-correct decision. In manual audits, a case is confirmed masked only after a reviewer first establishes $H(y,x) = 0$ from audio and then verifies surface recovery by at least one ASR route.

We distinguish two observable masking routes:

1. **Contextual recovery**: ASR uses sentence and entity context to prefer an intended score or model interpretation over the spoken relation marker.

2. **Canonicalization recovery**: ASR maps a letter-by-letter or otherwise noncanonical spoken realization to a conventional unit, product, or membership form.

The routes can overlap and cannot be identified solely from a transcript. We use the taxonomy, human-heard reading, and span-isolation behavior as diagnostic evidence rather than claiming direct access to internal decoder states.

4 Data and Human Audit Design

4.1 Source pool and frozen benchmark

We start from 108,124 company-produced Chinese news scripts used in a production TTS workflow. A scripted candidate-construction pipeline cleans titles and summaries, filters missing or unusable titles and rows without rule-detected auditable risk spans, scores candidates by the number and diversity of detected spans, and applies type/domain caps to select a 500-case real-news candidate pool. These filters use text and risk metadata before TTS or ASR generation; they are data-availability and coverage filters rather than outcome filters.

We additionally construct a 5K template-based synthetic hard-case pool to cover rare and controllable patterns. From the real and synthetic pools, we freeze a 200-case benchmark before human listening. The benchmark contains 155 real-news and 45 synthetic cases. It covers 85 CDRD-entity, 35 CDRD-polyphone, and 80 adjacent/non-CDRD cases across auto (44), general (43), finance (33), technology (31), sports (27), international (12), and military (10) domains. Each `case_id` occurs once.

Table 1: Risk-enriched construction and evaluation funnel.

Stage	Size	Purpose
Company-produced news source pool	108,124	taxonomy-driven risk mining
Real candidate pool	500	type/domain-capped real cases
Synthetic hard-case pool	5,000	rare and controlled risk coverage
Frozen benchmark	200	Raw/Structured human and automatic evaluation
Targeted audit pool	110	complete-denominator masking audit
Confirmed-masked audio files	97	46 MiMo plus 51 CosyVoice files; text cases can overlap

Most cases are grounded in real news and known product failure patterns, while risk mining, type caps, synthetic cases, and targeted selection deliberately enrich difficult spans.

4.2 The 200-case span audit

One primary annotator listened to Raw and Structured audio for all 200 frozen cases and counted correct risk spans. This is a targeted span-audit task, not a blind MOS or preference test. The review page contains the original text, Structured text, risk spans, expected and known negative readings, both audio conditions, and ASR transcripts.

The protocol is audio-first. The primary annotator and independent IAA annotators judge span correctness from audio against the reading references before consulting ASR transcripts as audit context. Transcripts are never treated as evidence that the audio is correct. A 30-case subset, stratified as 10 CDRD-entity, 10 CDRD-polyphone, and 10 adjacent/non-CDRD cases, is independently reviewed by two additional annotators.

For condition s, we report case-macro risk-span accuracy:

$$\mathrm{Acc}_s = \frac{1}{N}\sum_{i=1}^{N}\frac{n_{i,s}^{\text{correct}}}{n_i^{\text{risk}}}, \qquad s \in \{\mathsf{Raw}, \mathsf{Structured}\}. \tag{2}$$

If an entered count exceeds the number of spans, the official score clamps it to the total while preserving the original value for audit. We report exact correct-count agreement and Pearson correlation for the IAA subset. The primary annotation remains the main 200-case label set; IAA disagreements are not silently adjudicated into a new target.

4.3 The 110-case targeted audit

The targeted pool is mined from frozen high-risk candidates using CDRD/CDRD-adjacent surface rules, Raw/Structured disagreements, and suspected masking signals such as a surface-correct ASR transcript despite high-risk audio. A row enters as a *candidate*, not as a confirmed positive. Human outcomes are not used to construct the pool.

The primary reviewer first listens to Raw audio and decides whether the audited span is spoken incorrectly. Only then are ASR outputs consulted to assign one of five labels:

- **confirmed masked**: Raw TTS is wrong and at least one ASR route returns an intended or surface-correct form;
- **exposed TTS error**: Raw TTS is wrong and ASR exposes or preserves the wrong form;
- **no Raw TTS error**: Raw TTS is correct at the audited span;
- **uncertain**: evidence is insufficient for a stable decision;
- **not judgeable**: audio, span, reference, or metadata is unsuitable.

A second reviewer fully checks all 110 final categories. We also construct a 30-row label-blind relabel subset containing 15 confirmed-masked cases not re-exposed by span isolation, 7 exposed-error cases, and 8 no-error cases. This subset is intentionally weighted toward the most contestable confirmed cases and toward label boundaries; it is a robustness audit, not an independent random sample.

5 Systems and Evaluation Protocols

5.1 TTS conditions

The primary TTS audio is generated through the MiMo-V2.5-TTS API with a fixed voice (白桦) and WAV output. The same 110-case targeted pool is synthesized with CosyVoice-300M-SFT [23] using Raw input as a second-TTS validation. Edge TTS is retained only as an auxiliary automatic Raw/Structured control and does not enter the human masking evidence.

In the Raw condition, the TTS input retains the original news title and summary without rewriting risk spans into expected spoken forms. The same case-level pronunciation dictionary is supplied in both conditions; it can contain selected polyphone and foreign-name pronunciations, so Raw is a matched production baseline rather than a dictionary-free frontend. These entries do not implement the entity- and convention-span text rewrites that distinguish Structured. In the Structured condition, predefined risk annotations instantiate a spoken-text form by replacing written risk spans with expected readings. Negative readings and context evidence remain evaluation metadata and are not supplied to the TTS. Model settings are matched across conditions.

Because the Structured readings come from predefined rules and mappings that also define benchmark targets, Structured is an **oracle-style upper-bound diagnostic**. It tests whether explicit reading decisions reduce listener-perceived errors; it is not a deployable open-ended frontend and is not used as the paper's central method claim.

5.2 ASR systems

The reported MiMo transcripts are generated with the audio-capable `mimo-v2.5` API [24] under a strict transcription prompt that requests verbatim spoken forms, preserves spoken Chinese numerals, and discourages normalization. `mimo-v2-omni` serves as a fallback and protocol-ablation route. The prompt mainly constrains numeral rendering; it cannot guarantee that unit, product, or entity canonicalization is absent from decoder or post-processing behavior.

Whisper-small [15] is used as a non-MiMo control and for timestamp alignment. We further transcribe 220 human-audited Raw files (110 MiMo plus 110 CosyVoice) and the 46 aligned MiMo clips with two open-source ASR systems:

- **Paraformer-zh v2.0.4** [25], with FSMN-VAD v2.0.4, no punctuation model, hotwords, or external language model;
- **Qwen3-ASR-1.7B** [26], using its local Transformers backend with empty context and automatic language detection.

Neither open-source ASR receives source text, expected readings, negative readings, or a target-specific prompt. For Paraformer, toggling FunASR's `use_itn` flag changes none of the 220 full-file or 46 aligned-clip transcripts. We therefore treat it as a no-effect parameter check, not as evidence that ITN causes or prevents masking.

5.3 Automatic matching and protocol sensitivity

Automatic benchmark scoring checks whether an accepted expected reading is present and known negative readings are absent, subject to span-specific equivalences. Human-only spans are excluded. For targeted surface recovery, we use a conservative separator-normalized match against either the written target or an expected form; normalization removes whitespace, hyphens, colons, full-width colons, and middle dots. Occurrence-aware manual review resolves cases where a transcript contains both forms, neither predefined form, or multiple target occurrences.

This scoring is an audit aid rather than audio ground truth. We explicitly vary ASR protocol, and the manual targeted label requires a human-confirmed wrong reading before transcript recovery is considered.

5.4 Context-isolation diagnostic

For the 46 confirmed masked MiMo cases, at least one case-specific full-context ASR route returns an expected or surface-correct form. We produce two local views of each target span. A text-position heuristic extracts a rough six-second clip. Whisper-small timestamps then define an aligned clip: 35/46 use a candidate-string match and 11/46 use a nearest-chunk fallback. All isolated clips are transcribed with main strict MiMo ASR and manually categorized as exposed, still masked, no output, or other transcript.

This intervention removes linguistic context but can also change acoustic boundaries, silence, and decoder behavior. Because the full-context masking route varies by case whereas every isolated clip uses main strict MiMo ASR, the MiMo comparison is a cross-route mechanism diagnostic. Section 6.5 supplies a protocol-matched full-to-isolated comparison with Qwen3-ASR.

6 Results

6.1 Complete-denominator targeted audits

Table 2 is the primary denominator evidence. In the MiMo audit, human listening confirms 46 cases where Raw TTS is wrong and at least one ASR route returns an expected or surface-correct form. The same pool contains 9 exposed TTS errors and 55 cases with no Raw error. The CosyVoice audit uses the same 110 candidates and produces 51 confirmed masked, 27 exposed, 30 no-error, and 2 uncertain/not-judgeable cases.

Table 2: Targeted human-audit outcomes for two 110-case high-risk pools.

Outcome	MiMo Raw	CosyVoice Raw
confirmed masked	46	51
exposed TTS error	9	27
no Raw TTS error	55	30
uncertain / not judgeable	0	2
total targeted pool	110	110

For MiMo, Table 3 reports the outcome distribution by the frozen case-level benchmark label. A multi-risk case can ultimately be audited on a span from another family, so these labels are not a taxonomy of the final target spans. CDRD-entity supplies most confirmed cases; the type-capped selection process prevents population-level comparisons among rows.

Table 3: MiMo targeted outcomes by frozen case-level benchmark label. Columns are confirmed masked, exposed TTS error, and no Raw TTS error.

Label	Masked	Exposed	No error	Total
CDRD-entity	33	6	42	81
CDRD-polyphone	3	1	7	11
CDRD-adjacent / non-CDRD	10	2	6	18
Total	46	9	55	110

The 30-row blind relabel produces 23/30 exact full-label agreement ($\kappa = 0.634$). For the paper's primary binary distinction, confirmed masked versus other outcomes, agreement is 27/30 with $\kappa = 0.800$. The second reviewer also checks the full 110-row final set. These results do not eliminate judgment uncertainty, but they directly test the most consequential category boundary.

6.2 Representative cases and masking routes

Table 4 makes the false-negative event observable. The written target and ASR recovery alone can look correct; the error is in the spoken realization heard by a reviewer. The route column prevents a single-mechanism overclaim.

Table 4: Representative confirmed masked cases. Chinese strings in *Raw heard* record the listener-perceived reading.

Surface	Intended	Raw heard	ASR recovery	Route
13-11	十三比十一	十三至十一	score-correct form	contextual score
F-35	F 三五	F 杠三五	*F-35*	model-name recovery
伊尔-76	伊尔七六	伊尔负七六	aircraft-model form	entity recovery
88VIP	八十八 VIP	八十八伏 IP	*88VIP*	membership canonicalization
350Wh/kg	三百五十瓦时每公斤	partial unit-letter reading	normalized unit	unit canonicalization
640kW	六百四十千瓦	六百四十 kW	normalized unit	unit canonicalization

Target-span review of the 46 MiMo confirmed cases finds 20 sports score or formation spans, 10 aircraft or military-model spans, 15 technical-unit or torque spans, and one membership name. This span-level breakdown differs from the frozen case labels in Table 3 because several news items contain multiple risk families. The result is therefore not driven by one duplicated score template.

Protocol sensitivity further shows that the label is not an artifact of a single prompt. A conservative target/expected-form match identifies surface-correct recovery in 34/46 under default ASR and 27/46 under the MiMo-V2-Omni strict route. The final confirmed label requires human-confirmed wrong audio plus at least one surface-correct route.

6.3 Context isolation re-exposes local evidence

Table 5 reports the isolation diagnostic. Full sentences are masked by construction under their case-specific audit routes. Rough six-second clips expose 16 cases. Aligned clips expose 18, comprising 16 strong erroneous-reading recoveries and 2 partial unit-letter recoveries. Thirteen aligned clips yield no usable output.

Table 5: Context isolation on 46 confirmed MiMo cases. Full sentences use the original audit route; all clips use main strict ASR.

Condition	Exposed	Masked	No output	Other
Original full sentence	0	46	0	0
Rough six-second clip	16	11	17	2
Aligned clip, reviewed	18	12	13	3

Examples include *21-17* transcribed as 二十一至十七, *F-35* as F 杠三五, and 伊尔-76 as 伊尔负七十六 after isolation. These outputs preserve the local relation or model-marker reading that a full-context route had recovered to a surface-correct form. Because the MiMo full-context routes vary, the protocol-matched Qwen transition reported in Section 6.5 is the cleaner same-decoder control.

This pattern makes a purely acoustic explanation inadequate for the exposed subset: the ASR route can recover local wrong-reading evidence when sentence context is reduced. It does not prove that a particular hidden language-model state caused every transition, and the no-output rate prevents treating isolation as a complete metric.

6.4 Cross-TTS validation

CosyVoice is evaluated with Raw text on the same candidates, allowing a direct recurrence check across TTS systems. Thirty case rows are confirmed masked under both systems: 20 involve the same audited target span, while 10 involve different risk spans within the same multi-risk news item. Among the 51 CosyVoice confirmed cases, MiMo strict ASR returns a surface-correct form in 37, Whisper-small does so in 36, and both do so in 22. The overlap shows that the audit is not defined by a single ASR; the non-overlap shows evaluator sensitivity even within one TTS set.

CosyVoice also produces 27 exposed TTS errors, compared with 9 for MiMo. The systems misread different spans and realize some errors differently, so a fixed ASR can expose one acoustic form while canonicalizing another. Cross-TTS recurrence does not require identical error distributions.

6.5 Open-source cross-ASR controls

The two TTS audits contain 97 files previously labeled confirmed masked: 46 MiMo and 51 CosyVoice. They are a selected subset of all wrong-reading files, excluding errors already exposed by the audit ASR routes. Table 6 applies the same occurrence-aware surface-recovery criterion to Qwen3-ASR and Paraformer outputs.

Table 6: Occurrence-aware review on the 97 confirmed-masked files. “Preserved” includes wrong or noncanonical forms; “other” is neither surface recovery nor a clearly preserved wrong form.

ASR	MiMo rec.	Cosy rec.	Total rec.	Preserved	Other
Paraformer-zh	0/46	2/51	2/97	88/97	7/97
Qwen3-ASR-1.7B	19/46	21/51	40/97	56/97	1/97

Qwen reproduces substantial surface recovery outside the MiMo API. Among its 19 full-context MiMo recoveries, aligned-span transcription changes 12 to a wrong or noncanonical form, leaves 7 surface-correct, and creates no unpaired cases. Across all 46 paired MiMo files, aligned Qwen outputs contain 7 surface recoveries, 35 preserved wrong/noncanonical forms, and 4 other forms.

Paraformer provides a useful contrast. It surface-recovers only 2/97 confirmed-masked files and preserves a wrong or noncanonical form in 88/97. These outputs do not make Paraformer an audio ground truth; they show that masking strength depends heavily on decoder family and transcription protocol. The no-effect `use_itn` toggle also rules out a one-switch ITN explanation for this route.

6.6 Auxiliary 200-case Raw/Structured results

The Raw/Structured experiment asks whether making predefined reading decisions explicit reduces risk-span errors. Table 7 reports paired case-macro accuracy and bootstrap confidence intervals. Structured improves overall accuracy by 0.0614, with the largest change on CDRD-entity cases. Polyphone performance is near ceiling in both conditions.

Table 7: Human case-macro risk-span accuracy. Structured is an oracle-style diagnostic, not a deployable-system comparison.

Split	N	Raw	Structured	Δ (95% CI)
Overall	200	0.8889	0.9503	$+0.0614$ [0.0352, 0.0891]
CDRD-entity	85	0.7887	0.9146	$+0.1259$ [0.0728, 0.1807]
CDRD-polyphone	35	0.9914	0.9854	-0.0060 [-0.0204, 0.0057]
Adjacent / non-CDRD	80	0.9505	0.9729	$+0.0224$ [-0.0024, 0.0560]

On the 30-case IAA subset, pairwise exact agreement for correct counts ranges from 0.900 to 0.933 on Raw and 0.833 to 0.900 on Structured; Pearson correlations range from 0.848 to 0.948. These results support the span-count protocol while also showing that Structured judgments are not mechanically unanimous.

ASR scores are directionally useful but protocol-sensitive. Raw/Structured automatic accuracy ranges from 0.495/0.528 under default ASR to 0.728/0.805 under MiMo-V2-Omni strict ASR. Under the main strict route, the values are 0.6121/0.7826. On the aligned 196-case auto-valid subset, human labels exceed strict-ASR scores by 0.2745 for Raw (95% CI [0.2184, 0.3320]) and 0.1667 for Structured (95% CI [0.1146, 0.2220]). This calibration gap and the targeted audit answer different questions: the former concerns aggregate score calibration, while the latter establishes a specific false-negative event.

7 Analysis and Implications

7.1 What the evidence establishes

The audits identify a concrete failure: a listener hears the wrong target-span reading while the transcript appears correct. It recurs across several news-risk families and two TTS systems. Reducing context exposes local wrong-reading evidence in part of the MiMo set, and Qwen reproduces the same full-to-isolated transition under a matched decoder. Paraformer mostly preserves the wrong or noncanonical reading instead.

The claim is narrower than a general mismatch between ASR metrics and human judgments. It concerns a directed false negative whose frequency changes with decoder family, prompt, normalization behavior, and available context.

7.2 Contextual recovery versus canonicalization

Scores and model names provide the clearest contextual-recovery examples. The spoken marker in 十三至十一 or F 杠三五 is locally audible, but sentence context strongly favors a score or aircraft interpretation. Units and membership strings can follow a different route: the ASR output may canonicalize a partial letter reading into a conventional written token. The strict MiMo prompt discourages normalization, especially of numerals, but it cannot fully control unit- or entity-level canonicalization. This explains why a "strict" protocol can still produce normalized recoveries without contradicting its intended constraint.

The distinction affects interpretation. Span isolation is strongest evidence for context-sensitive recovery when it changes a full-context surface recovery into the human-heard wrong form. It is weaker evidence for cases where isolation still returns a canonical unit or no output. We therefore report routes and transition counts rather than assigning every case to one latent mechanism.

7.3 Recommended evaluation practice

For TTS benchmarks enriched with short context- or convention-dependent readings, we recommend:

1. define target spans, accepted readings, and known negative readings before scoring;
2. use audio-first human audit for high-risk spans and do not treat an ASR transcript as evidence of audio correctness;
3. report the full targeted denominator, including exposed errors and no-error candidates;
4. compare ASR systems from different decoder families rather than relying on one prompted ASR-LLM;
5. use context isolation only as a diagnostic, with protocol-matched subsets and no-output outcomes reported;

6. separate prevalence claims from hard-case mechanism claims.

ASR roundtrip remains valuable for scalable screening, regression testing, and protocol ablation. It should be calibrated against human listening rather than used as standalone ground truth through a single transcript route.

7.4 Role of the Structured intervention

Structured supplies expected readings from the same rule and mapping resources that define benchmark targets. Its 0.0614 gain is therefore an upper-bound diagnostic: it helps localize the failures to reading decisions, but it is not a fair competition between deployable frontends.

8 Limitations

The 110-case audit is rule-selected and signal-enriched. It establishes existence and mechanism; estimating production incidence requires a random or stratified production sample.

The primary 200-case labels and initial 110-case labels are each produced by one primary annotator. Reliability is strengthened by two independent annotators on the 30-case span-count subset, a complete second review of the 110 categories, and an additional label-blind 30-case audit, but larger independent annotation would further reduce uncertainty.

The listening tasks are reference-guided rather than naive-listener detection tests. They establish whether a target span follows the specified news reading, not how often an unaided listener would spontaneously notice the error.

CosyVoice reuses the same candidates and is Raw-only, which supports cross-TTS recurrence but not an independent distribution estimate. Edge remains an automatic-only control. Broader independently sampled TTS pools would improve external validity.

Context isolation changes both linguistic and acoustic context. Timestamp alignment can truncate neighboring phones, and 13/46 aligned MiMo clips produce no usable output. The diagnostic therefore supports a mechanism interpretation for exposed transitions but is not a replacement scoring method.

The Qwen3-ASR and Paraformer controls demonstrate evaluator dependence; they do not cover the full range of ASR architectures. Paraformer's 2/97 recovery count should not be interpreted as zero masking in all conventional ASR systems, and the unchanged `use_itn` outputs do not identify ITN as a causal factor.

Finally, CDRD-adjacent categories combine multiple convention-dependent phenomena. We make this boundary explicit and report strict CDRD and adjacent counts separately where the data support it. Future work should expand risk discovery, add random-sample incidence estimation, and evaluate more open-source TTS and ASR families.

9 Reproducibility, Data Availability, and Ethics

Supporting materials for this work are released through the project GitHub repository. The full audio-and-data archive is deposited on Zenodo under DOI 10.5281/zenodo.21454402. Together, the release contains frozen case metadata, risk-span annotations, prompts, API model identifiers, ASR transcripts, scoring scripts, de-identified human labels, targeted-audit tables, context-isolation summaries, the Paraformer/Qwen occurrence-aware audits, and released audio. The public text package contains 500 company-authorized production news scripts and 5,000 synthetic hard cases; the full 108,124-item source export is excluded.

The reported MiMo transcripts were generated with the MiMo `mimo-v2.5` API under the released strict prompt. The API model identifiers, prompts, settings, outputs, and scoring scripts support audit and API-based reruns. Released code is licensed under the MIT License; the released company-authorized production news scripts, synthetic cases, generated audio, annotations, transcripts, and derived metadata are licensed under CC BY 4.0. Annotator identities are not released.

This study uses news text, synthesized speech, ASR transcripts, and human listening judgments for TTS evaluation. Annotators performed span-level listening audits of synthetic audio and authorized the use of their anonymized judgments. No sensitive personal data, medical data, biometric identification, speaker identification, or clinical intervention is involved. The study follows the principles of the Declaration of Helsinki. It was reviewed under the authors' institutional legal and compliance process and was determined not to require formal ethics-board review.

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10 Conclusion

ASR roundtrip can hide listener-perceived reading errors in Chinese news TTS by returning a surface-correct transcript for wrong audio. The targeted MiMo audit contains 46 masked, 9 exposed, and 55 no-error cases. MiMo context isolation re-exposes 18/46, and the matched Qwen comparison changes 12/19 full-context recoveries to wrong or noncanonical forms on aligned spans. CosyVoice reproduces the failure family, while Qwen3-ASR (40/97) and Paraformer (2/97) differ sharply in surface recovery on the confirmed-masked subset. ASR roundtrip remains useful for screening, but reading-risk evaluation needs human listening anchors, span-level references, and protocol-diverse controls.

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A Detailed Agreement Results

A.1 Thirty-case Raw/Structured IAA

Table 8 reports the three pairwise comparisons among the primary annotator and two independent annotators. Exact agreement is computed on the number of correct risk spans per case after the documented clamp rule.

Table 8: Pairwise agreement on the 30-case Raw/Structured span-count subset.

Pair	Raw exact	Structured exact	Raw r	Structured r
Primary vs reviewer 2	0.9000	0.8667	0.9099	0.9241
Primary vs reviewer 3	0.9000	0.8333	0.8480	0.9066
Reviewer 2 vs reviewer 3	0.9333	0.9000	0.9443	0.9478

A.2 Blind targeted-audit relabel

Table 9 gives the full-label confusion matrix for the label-blind 30-case relabel. Original rows were selected as 15 confirmed masked, 7 exposed error, and 8 no error; the blind reviewer assigned 14 confirmed masked, 4 exposed error, 11 no error, and 1 uncertain.

Table 9: Original targeted label versus blind relabel.

Original \ blind	Masked	Exposed	No error	Uncertain
confirmed masked	13	2	0	0
exposed TTS error	1	2	3	1
no Raw TTS error	0	0	8	0

Full-label agreement is 23/30 (0.767; $\kappa = 0.634$). Excluding the one uncertain row, three-way agreement is 23/29 (0.793; $\kappa = 0.665$). For confirmed masked versus all other outcomes, agreement is 27/30 (0.900; $\kappa = 0.800$). Using the blind reviewer's explicit Raw-span correctness field, Raw-wrong versus Raw-correct agreement is 27/30 (0.900; $\kappa = 0.772$).

B Additional Automatic Results

B.1 ASR protocol sensitivity

Table 10 summarizes the Raw/Structured protocol ablation.

Table 10: Automatic case-macro accuracy under MiMo ASR protocols on Raw and Structured audio.

ASR protocol	Raw	Structured
Default ASR	0.495	0.528
Main strict prompt	0.612	0.783
Strict plus negative readings	0.612	0.783
MiMo-V2-Omni strict	0.728	0.805

The unchanged strict-plus-negative result indicates that the additional negative-reading text did not alter this ASR route on the benchmark. The large absolute movement across protocols is a reminder that automatic scores are properties of the TTS/ASR/protocol combination.

B.2 Human-ASR calibration by category

Table 11: Human minus strict-ASR case-macro accuracy on the aligned 196-case subset.

Label	Condition	N	Human	ASR	Gap (95% CI)
Overall	Raw	196	0.8866	0.6121	0.2745 [0.2184, 0.3320]
Overall	Structured	196	0.9493	0.7826	0.1667 [0.1146, 0.2220]
CDRD-entity	Raw	85	0.7887	0.5697	0.2190 [0.1542, 0.2852]
CDRD-entity	Structured	85	0.9146	0.7582	0.1564 [0.0694, 0.2461]
CDRD-polyphone	Raw	35	0.9914	0.7333	0.2582 [0.1258, 0.4036]
CDRD-polyphone	Structured	35	0.9854	0.8211	0.1644 [0.0638, 0.2773]
Adjacent	Raw	76	0.9479	0.6038	0.3441 [0.2360, 0.4567]
Adjacent	Structured	76	0.9715	0.7922	0.1793 [0.0974, 0.2690]

Four foreign-name-only cases have no auto-valid span after human-only exclusions, yielding the aligned $N = 196$ subset. The gap is a calibration comparison and must not be interpreted as the proportion of masked cases.

B.3 Auxiliary evaluator and TTS controls

Table 12: Auxiliary Raw/Structured automatic controls. These do not replace the human-audited masking evidence.

Setup	Raw	Structured	Δ
Edge TTS plus strict ASR	0.7219	0.8609	+0.1390
Whisper-tiny ASR	0.4700	0.5398	+0.0698
Whisper-small ASR	0.5332	0.5994	+0.0662

On an older 50-case four-pipeline set, TTS-PRISM pronunciation scores are 4.50 for Raw, 4.44 for Prompt, 4.62 for a PolyNorm-style condition, and 4.44 for Structured. This limited negative control does not characterize all learned evaluators; it shows that one broad pronunciation score does not track the human reading-decision gain.

C Protocol Details

C.1 Audio-first audit order

The audit order is designed to prevent ASR output from becoming the primary perceptual evidence. Annotators first read the source sentence, identify the highlighted risk span, and inspect the expected and known negative readings. They then listen to the relevant TTS audio and record span correctness. Only after this audio decision do they inspect ASR transcripts to classify masking or exposure. They record uncertainty rather than force a category when audio or references are inadequate.

For the 200-case Raw/Structured task, annotators count correct spans rather than choose a preferred system. For the 110-case audit, the first decision is Raw audio correctness; the second decision is whether an ASR route masks or exposes the human-heard error.

C.2 Strict transcription instruction

The released protocol asks the MiMo ASR API to transcribe the audio verbatim, preserve how numerals and symbols are spoken, avoid rewriting a heard reading into a more plausible written form, and return only the transcript. This instruction primarily targets numeral and separator behavior. The reported outputs show that prompting changes aggregate scores but does not eliminate contextual or canonicalizing recovery.

C.3 Occurrence-aware surface recovery

The Qwen and Paraformer audits operate on the target occurrence, not on document-level string presence alone. Reviewers inspect the transcript around the human-confirmed wrong occurrence and assign one of three relations: exact surface-correct recovery, wrong/noncanonical form preserved, or other without exact recovery. This avoids crediting an unrelated correct occurrence elsewhere in a long news item.